A Symmetric Layer–Union Audit of Component Collapse in Hierarchical Procedural Corpora

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Abstract

Component-disjoint leakage control can group corpus units by content similarity, hierarchical membership, or both. Guvenilir and Doğan previously showed that merging relation types can create a giant component that obstructs splitting; we do not claim this phenomenon as new. We examine it through a symmetric audit of a content-near-duplicate layer, a common-container layer, and their union in a fixed panel of six hierarchical procedural corpora. MyFixit and Doc2Dial exhibit the individual-layer-pass/union-fail pattern under the same operational criteria. The resulting two-of-six fraction describes this deliberately constructed panel and is not a prevalence estimate. A prespecified bridge-specific predictor is associated with the pattern, but it is not distinguished from a registered union-density control; the panel therefore does not identify a bridge-specific mechanism. Secondary diagnostics bound the interpretation of threshold sensitivity, annotation coverage, and lexical cues without extending those findings beyond their recorded sources and definitions. The paper's contribution is a bounded measurement and audit: it keeps relation families visible, evaluates their individual and union component structures symmetrically, and reports negative cases and mechanism limits. It proposes neither a new splitting algorithm nor a general causal claim about relation unions.

1 Introduction

Leakage control often begins by deciding which data items must not be split across training and evaluation partitions. In a hierarchical procedural corpus, multiple relations can justify grouping. Procedural steps may express the same or near-duplicate content, and otherwise different steps may belong to the same guide, article, document, paragraph, or protocol. Either relation can be represented as a graph whose connected components are kept intact. If the relations are merged before transitive closure, however, a short path that alternates between relation types can join units that neither layer connects on its own. The resulting quarantine groups may therefore have very different capacity from those suggested by either layer separately.

This interaction sits within a mature literature on similarity-aware and leakage-reduced splitting. Lo-Hi, GraphPart, DataSAIL, PLINDER, and related work construct graph- or cluster-based partitions and document the tension between separation and data retention [12, 13, 8, 4]. The closest direct precedent identified in our bounded review is Guvenilir and Doğan's drug–target-interaction study [7]. That work combines multiple relation types, reports a giant component that

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prevents component-disjoint splitting, and applies deletion and community-based repair. We therefore do not claim the broad chain from merged relations to split infeasibility as a new observation. Record-linkage work has likewise long noted that an erroneous link can propagate through transitive closure [6], and near-duplicate handling is an established concern in information-retrieval evaluation [5].

The narrower question studied here is a cross-domain measurement question. When hierarchical procedural corpora provide both natural-language content and an explicit document-like container, how do the component summaries of a content-near-duplicate layer, a common-container layer, and their union compare under a symmetric operational audit? The emphasis is on retaining the identity of each edge family. A large union component is informative only when it can be compared with each constituent configuration under the same unit mapping, closure rule, summaries, and decision criteria.

We evaluate a fixed panel of 6 gate-eligible sources drawn from several procedural domains and provenance families. For each source, Section 2 maps source records to procedural units, hierarchical containers, and content fields. C2 links normalized exact and near-duplicate content, C3 links membership in the same container, and C5 is the transitive closure of their union. Component count, largest-component share, and effective component count describe the resulting partitions. The source-level rule asks whether both individual layers retain sufficient component dispersion under criteria A1–A4 while their union fails the same checks. MyFixit was already known to meet this pattern; 5 additional candidates were screened, so the panel is outcome-enriched rather than a basis for population inference.

The measurement yields 2 qualifiers among the 6 eligible sources. MyFixit and Doc2Dial, drawn from 2 provenance families, show the individual-pass/union-fail pattern. The other sources remain visible as negative cases. Accordingly, 2/6 is reported only as a descriptive panel fraction, not as a base rate or prevalence estimate.

The mechanism-discrimination criterion is not met. Although a bridge-specific predictor tracks the qualifier pattern, the registered union-density control satisfies the same separation and outcome-association conditions. The observed configuration contrast therefore does not identify a bridge-specific mechanism. The study retains this negative result rather than using secondary diagnostics to rescue a stronger explanation.

The study contributes a symmetric layer-versus-union measurement, an explicit mechanism-discrimination test, and tightly scoped secondary diagnostics. It reports the source mapping and nonqualifying cases, states the operational criteria in auditable component summaries, and preserves the narrow threshold-grid, coverage-exposure, and lexical findings within their recorded evidence domains. These contributions do not include a splitting algorithm, a giant-component theorem, an entity-resolution method, or a causal account of bridging.

The rest of the paper proceeds as follows. Section 2 defines the panel, typed layers, component summaries, and decision rules. Sections 3 and 4 report the source-level measurement and audit-rule checks. Sections 5 and 6 present the mechanism and threshold boundaries. Sections 7 and 8 report the diagnostic analyses. Sections 9– 11 position and bound the evidence before Section 12 concludes.

2 Measurement design and operational definitions

2.1 Panel and source-specific units

The measurement treats each corpus as a collection of procedural units nested inside source-defined containers. A unit is the smallest occurrence to which the source supplies usable procedural text; a container is the document-like object whose members should remain together under the structural relation. The mapping is source-specific because the same abstract roles are represented by different schemas. In Human Know-How, a unit is a label-bearing target of a structured `has_step` relation and its container is the parent WikiHow article. In MyFixit, guide-step occurrences are nested in repair guides. OpenPI process-step records are mapped to their WikiHow articles; WIQA paragraph steps to ProPara procedure paragraphs; Doc2Dial non-heading document spans to government-service documents; and X-WLP operation events to wet-lab protocols. The corresponding label, step, span, or sentence field is the natural-language content field used by the content relation.

The fixed mapping yields 6 gate-eligible sources: Human Know-How, MyFixit, OpenPI, WIQA, Doc2Dial, and X-WLP. Eligibility requires both a resolvable hierarchical container and a natural-language content field from which the two relation families below can be defined. WHPG is excluded because one malformed record prevents an article identifier from being resolved for all of its units. COIN is retained only as an auxiliary diagnostic outside the gate-eligible panel because its segment label is a closed-vocabulary taxonomy item rather than natural-language procedural text. The panel is not a probability sample. MyFixit entered as the previously known qualifying source, and 5 additional candidates were screened. Results are interpreted as outcomes for this fixed panel, not as estimates of population prevalence.

Panel composition for the six gate-eligible sources. The following unnumbered summary describes only panel size and container composition; it does not display qualifier status or anticipate the outcome. Means are computed from the full-precision unit and container counts. The median column is included because all six frozen aggregate diagnostics contain a non-null value.

Source	Units	Containers	Units per container		
			Mean	Median	Maximum
Human Know-How	80,286	9,967	8.06	6	206
MyFixit	235,549	18,995	12.40	9	137
OpenPI	4,050	810	5.00	5	6
WIQA	2,557	379	6.75	6	10
Doc2Dial	31,602	488	64.76	56	494
X-WLP	3,915	279	14.03	14	36

Eligibility is structural rather than a claim of genre homogeneity. Doc2Dial provides task-oriented government-service documents, while the other panel members include repair guides, instructional articles, procedure paragraphs, and laboratory protocols. Doc2Dial satisfies the same formal requirements for a unit, container, and content field. This subject-matter heterogeneity limits external interpretation of the fixed panel but does not change source eligibility.

2.2 Typed relation layers and closure

For each source, let V be its mapped unit occurrences. We construct two undirected edge families on the common vertex set V. The content layer (C2) links units whose normalized content is identical or satisfies the registered near-duplicate rule. Normalization case-folds Unicode text, replaces punctuation and underscores with spaces, and collapses whitespace; the non-exact comparison uses word-trigram sets and registered Jaccard similarity. Very short strings connect only under normalized exact equality. The container layer (C3) expands every source-defined container into a clique, linking every pair of its units. Thus C2 represents reusable or repeated content, whereas C3 represents membership in one article, guide, paragraph, document, or protocol. The labels describe graph construction and do not assert that every edge is a semantic equivalence or a causal dependency.

The union configuration C5 contains both C2 and C3 edges. In every configuration, quarantine groups are the connected components obtained by transitive closure: if any path connects two units, they belong to the same indivisible group for a component-disjoint split. The comparison is symmetric. C2, C3, and C5 are evaluated with the same component summaries and the same operational rule. This design isolates an observed configuration contrast; it does not treat the union as the only legitimate representation or infer a general causal effect from that contrast.

2.3 Component summaries and the source-level rule

Suppose a configuration partitions N units into components of sizes $n_1, \ldots, n_m$, and write $p_j = n_j/N$. We report the component count m, the largest-component share $\max_j p_j$, and the effective component count

$$N_{\text{eff}} = \left(\sum_{j=1}^{m} p_j^2 \right)^{-1}.$$

Component count records how many indivisible groups remain, largest-component share captures the largest immediate capacity constraint, and effective component count summarizes concentration across the full component-size distribution. These quantities are structural diagnostics; the study does not construct or optimize a new train–test split.

The source-level rule applies criteria A1–A4 to C2, C3, and C5. Criterion A1 requires largest-component share at most 0.01 and is a prespecified stress rule rather than a community-standard cutoff. Criterion A2 requires that share to be at most 0.20, the necessary fit bound for an indivisible component in the specified split geometry. Criterion A3 requires at least 30 effective components, incorporating the specified reserve above the minimum number of groups. Criterion A4 requires largest-component share at most 1/3, the equal-fold fit bound at the specified fold count. As Section 4 makes explicit, A1, A2, and A4 reuse the same statistic and are neither independent tests nor a general theorem of split feasibility.

A source qualifies when both individual layers pass at least 3/4 criteria and their union fails at least 3/4. Gate 1 asks whether at least 2 gate-eligible sources meet this source-level rule. Because provenance can be shared across derived datasets, outcomes are reported both by source and by provenance family.

2.4 Mechanism-discrimination and sensitivity boundaries

Gate 2 asks whether a prespecified bridge-specific predictor explains the qualifying pattern beyond simple graph density. The selected candidate, `bridge_edge_density`, counts cross-container content edges per container. It must separate qualifiers from nonqualifiers in the registered direction, reach the registered association threshold with C5 largest-component share, and remain separated in the one-source-deletion audit. Registered naive predictors are then tested as negative controls. In particular, `mean_union_degree` measures ordinary C5 edge density. The mechanism-discrimination requirement is not met if such a control also satisfies the separation and outcome-association conditions. A repeated source-level pattern and a bridge-specific explanation are therefore distinct claims; establishing the former does not establish the latter.

The remaining analyses preserve narrower evidence domains. The registered τ sweep concerns a discrete grid and a multiplicity-adjusted winner-flip decision, not an equivalence test. The coverage-exposure and lexical audits concern only the corpora and definitions represented in their source

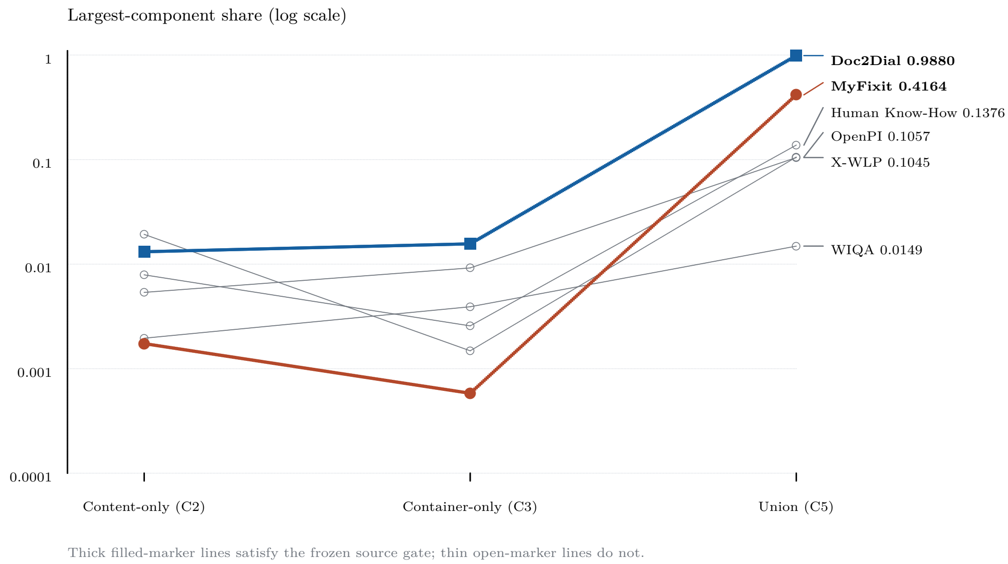

Figure 1: Largest-component share under the two individual layers and their union for all 6 gate-eligible sources. The vertical axis is logarithmic. Bold colored lines identify the 2 sources satisfying the frozen source gate; gray lines identify nonqualifying sources. The highlighting records an audit outcome, not a causal explanation.

records. These secondary checks bound interpretation of the main configuration measurement; they do not enlarge the source-level rule or the population to which its result applies.

3 Union-induced collapse appears in 2 of 6 gate-eligible sources

Doc2Dial provides the clearest observed instance by collapse magnitude. The largest connected component contains 1.31% of units under the content-only layer and 1.56% under the container-only layer. Under transitive closure of their union, the corresponding share is 98.80%. Only 18 components remain, and the effective number of components falls to 1.024.

MyFixit supplies a second, less extreme instance from a different provenance family. Its content-only and container-only largest-component shares are 0.17% and 0.06%, respectively; the union share is 41.64%. The union contains 5,375 components and has an effective component count of 5.760.

Under the audit criteria defined in Section 4, MyFixit is the only qualifying source whose individual layers both pass all 4 registered criteria. Each Doc2Dial individual layer fails only criterion A1 and passes criteria A2–A4.

2 of the 6 gate-eligible sources in the frozen panel satisfy the source Gate. This 2/6 value is a descriptive panel fraction. It is not an estimate from independent trials or a prevalence estimate. MyFixit was the previously known qualifying source. Among 5 additionally screened candidates, Doc2Dial was the sole qualifier (1 of 5). The 2 qualifying sources belong to 2 distinct provenance families, `DOC2DIAL_GOV` and `IFIXIT`.

WIQA provides a useful negative case: its union share is 1.49%, only modestly above the 0.20% and 0.39% shares of its individual layers, and its content layer contains only 63 edges. Union formation alone therefore does not entail collapse; cross-container content repetition must be present in the measured graph.

Table 1: Closure metrics for the gate-eligible sources. C2 and C5 use the frozen near-duplicate threshold $\tau = 0.85$; C3 uses exact container identity. $N_{\text{eff}} = (\sum_j p_j^2)^{-1}$ for component unit shares p_j.

Source	Layer	Components	Largest share	N_{eff}
Doc2Dial	Content-only (C2)	21,476	0.013132	2287.732
	Container-only (C3)	488	0.015632	325.232
	Union (C5)	18	0.988039	1.024
MyFixit	Content-only (C2)	84,777	0.001736	12870.890
	Container-only (C3)	18,995	0.000582	10209.086
	Union (C5)	5,375	0.416427	5.760
Human Know-How	Content-only (C2)	77,453	0.007897	12922.284
	Container-only (C3)	9,967	0.002566	6014.607
	Union (C5)	8,550	0.137558	52.215
OpenPI	Content-only (C2)	3,908	0.019259	1533.517
	Container-only (C3)	810	0.001481	788.961
	Union (C5)	683	0.105679	74.520
WIQA	Content-only (C2)	2,508	0.001955	2436.917
	Container-only (C3)	379	0.003911	359.541
	Union (C5)	341	0.014861	267.337
X-WLP	Content-only (C2)	2,213	0.005364	1422.480
	Container-only (C3)	279	0.009195	226.169
	Union (C5)	180	0.104470	49.318

Table 1 reports the content-only, container-only, and union configurations for every gate-eligible source. The largest-component share captures the most visible concentration, whereas the effective component count summarizes the full component-size distribution. In Doc2Dial, the union is close to a single-component partition by both summaries; in MyFixit the concentration is substantial but not complete. The 4 nonqualifying sources remain in the table to make the 2/6 denominator and the negative cases explicit.

4 Audit criteria and observed rule-shape stability

The source-level rule comprises criteria A1–A4. The content-only and container-only layers must each pass at least 3/4 criteria, while the union layer must fail at least 3/4. This is the rule under which the 2/6 result was obtained. Its thresholds, source eligibility, and outcome are unchanged in the analysis below.

The criteria are operational components rather than independent statistical tests. A1 is a prespecified stress criterion requiring largest-component share at most 0.01. A2 applies the same statistic at the specified smallest-split threshold 0.20; its grounding is the necessary fit of an indivisible component. Lo-Hi illustrates that rationale by assigning connected components independently while reporting that a giant component prevents its required split; it does not supply the present A2 threshold [12]. A3 instead requires at least 30 effective components. Its grounding combines the GroupKFold requirement for at least the specified 3 groups with a 10-fold reserve; that reserve is a stress margin, not an external theorem. A4 returns to largest-component share and requires at most 1/3, the equal-fold fit threshold at the specified fold count. Once the fold count k is fixed, A4 introduces no additional threshold-tuning degree of freedom. Thus A1, A2, and A4 are nested thresholds on one statistic, whereas A3 uses effective components. Counting criteria therefore repeats weight on largest-component share.

In the observed 6-source panel, every union layer fails A1; all 6 union outcomes have that status. Consequently, within this panel only, the union requirement is equivalent to failing at least

2 further criteria among A2–A4. This is a panel-specific equivalence in the observed outcomes, not a permanent rewrite of the decision rule.

Table 2: Independently recomputed union-layer criterion outcomes within the frozen 6-source panel. The failure count uses A2–A4 only; qualifying status remains the original A1–A4 source Gate.

Source	A1	A2	A3	A4	A2–A4 failures	Frozen qualifier
Human Know-How	FAIL	PASS	PASS	PASS	0/3	no
MyFixit	FAIL	FAIL	FAIL	FAIL	3/3	yes
OpenPI	FAIL	PASS	PASS	PASS	0/3	no
WIQA	FAIL	PASS	PASS	PASS	0/3	no
Doc2Dial	FAIL	FAIL	FAIL	FAIL	3/3	yes
X-WLP	FAIL	PASS	PASS	PASS	0/3	no

Table 2 shows the independently recomputed union-layer matrix. Within this 6-source panel, the 2 qualifiers have the 3/3 failure pattern on A2–A4, whereas all 4 nonqualifiers have the 0/3 pattern. No source has an intermediate failure count; the observed count of such sources is 0. This complete separation is consistent with the 2/6 qualifying set; it is not a significance test, predictive validation, or an additional source of replication evidence.

As a rule-shape check, all 3 nonzero failure-count cutoffs 1/3, 2/3, and 3/3 each select the same sources as the source-level rule. The 3 single-criterion checks using A2, A3, or A4 also select that same set within this panel. The A4 result is the cleanest single-criterion check here because, after the fold count is fixed, it adds no further threshold choice. These checks do not show that criterion choice is insensitive outside this panel.

This A2–A4 analysis is post hoc, presentation-only, and descriptive. It neither changes the original A1–A4 rule nor defines a new prespecified decision rule. The separation does not establish a mechanism and is not evidence about prevalence or behavior outside the fixed panel.

5 The bridging mechanism is not identifiable in the six-source panel

The prespecified bridging predictor, `bridge_edge_density`, shows a strong association in the fixed panel: it perfectly separates the 2 qualifying sources from the remaining sources, meets the registered correlation condition, and has a strictly separated full-panel range (minimum qualifier 106.648 versus maximum nonqualifier 20.891). Nevertheless, the mechanism-discrimination criterion is not met. The registered naive control, `mean_union_degree`, completely separates the two qualifiers from the four nonqualifiers. Its Spearman correlation with C5 largest-component share is $\rho_s = 0.829$, meeting the registered $\rho_s \geq 0.8$ condition. The negative control therefore satisfies both the separation and outcome-correlation conditions.

Under the frozen Gate-2 rule, this result is attributable to lack of discriminant validity against the registered negative control, not to absence of an observed bridge association in this six-source panel. Across the 6 sources, the predictor vectors have Spearman rank correlation $\rho_s = 0.943$. Their descending ranks agree for every source except OpenPI and WIQA, which exchange adjacent positions; there is only 1 discordant source pair. Both predictors place Doc2Dial and MyFixit first and second, so the observed binary outcome cannot distinguish bridge-specific structure from simple union-graph density.

Table 3: Frozen predictor values and within-panel descending ranks. The two predictors differ only in the OpenPI/WIQA ordering.

Source	Qualifies	Bridge density (rank)	Mean union degree (rank)
Human Know-How	no	20.891 (3)	17.536 (3)
MyFixit	yes	106.648 (2)	39.273 (2)
OpenPI	no	4.100 (5)	5.773 (6)
WIQA	no	0.161 (6)	6.160 (5)
Doc2Dial	yes	405.340 (1)	108.686 (1)
X-WLP	no	6.864 (4)	17.288 (4)

Table 4 records both predictors' frozen separation thresholds, slacks, and correlations with C5 largest-component share. Three correlations must be kept distinct: bridge-to-C5 is 0.943; negative-control-to-C5 is 0.829; and the cross-predictor correlation is 0.943. The negative-control-to-C5 value meets the registered $\rho_s \geq 0.8$ condition. None of the differences between these outcome correlations, thresholds, or slacks is credited in favor of the bridge predictor. Reporting the values is an audit record, not a post hoc comparison that rescues the mechanism.

Table 4: Frozen separation records for the selected bridge predictor and the registered naive control. Threshold, slack, and outcome correlation are reported for audit completeness; under the frozen negative-control rule, their numerical differences do not support the bridge predictor.

Predictor	Separates	Threshold	Slack	ρ_s with C5 share
`bridge_edge_density`	yes	63.7698	42.8783	0.943
`mean_union_degree`	yes	28.4043	10.8687	0.829

Table 5 is a one-source-deletion separability audit. For each row, the threshold is recomputed on the retained sources and the omitted source is not classified; this is not predictive cross-validation. Because full-panel separation is strict ($106.648 > 20.891$) and both classes remain after every single deletion, persistence under all 6 deletions is mechanically implied. The table verifies the arithmetic of that implication but supplies no independent robustness evidence and cannot change the mechanism conclusion.

Table 5: One-source-deletion separability audit for the registered bridge predictor. For each row, the threshold is recomputed on the retained sources and the omitted source is not classified. This is not predictive cross-validation. Because the full panel is strictly separated and both classes remain after every deletion, persistence is mechanically implied and supplies no independent robustness evidence.

Omitted source	Separates	Threshold	Slack
Human Know-How	yes	56.7560	49.8922
MyFixit	yes	213.1158	192.2244
OpenPI	yes	63.7698	42.8783
WIQA	yes	63.7698	42.8783
Doc2Dial	yes	63.7698	42.8783
X-WLP	yes	63.7698	42.8783

A key limitation of the mechanism design is that predictor distinguishability was not assessed before the Gate-2 decision rule was frozen. Freezing the rule before outcome interpretation prevented a post hoc rescue, but it did not make the control orthogonal. The analysis shows that unique explanatory gain was not demonstrated; it cannot decide which of two nearly aligned predictor families better accounts for the observed association. A future result-blind predictor-geometry audit should precede outcome testing, and the source panel should contain cases where the candidate mechanism and simple graph density make different predictions.

The observed correlation shows that the predictor rankings are nearly aligned in this six-source panel, but no defensible numerical panel-size target can be derived. Without a source-sampling design, a fitted outcome model, or a defined precision criterion, the required number of sources remains unidentified.

The density control also depends on how container connectivity is represented. C3 expands each container into a clique, so connected components depend on container membership but `mean_union_degree` depends on the edge expansion. A star or spanning forest can preserve the same within-container connectivity with fewer edges. In a post hoc, panel-specific sensitivity analysis using only frozen aggregate counts, we set $T_s = n_s - c_s$ and $O_s = E_{2,s} + E_{3,s} - E_{5,s}$ and bound the unknown overlap x_s between C2 and forest edges by $\max(0, T_s - (E_{3,s} - O_s)) \leq x_s \leq \min(T_s, O_s)$; the corresponding union edge count is $E_{2,s} + T_s - x_s$. The conservative aggregate-compatible mean-degree intervals leave the lowest qualifier value (14.49) above the highest nonqualifier value (6.96). Enumerating every compatible weak rank order, with average ranks for permitted ties, gives correlations with C5 largest-component share from 0.886 to 0.943, all above the registered 0.8 threshold. These bounds do not identify a unique spanning tree, and their endpoints need not both be attained by a constructible forest; graph-connectivity constraints could narrow them. No raw graph was read or concrete forest constructed. The original control remains clique-based, mean degree is not representation-invariant, and the frozen Gate-2 result is unchanged.

6 The registered τ grid is decision-inert, not metric-invariant

The frozen sweep artifact covers 2 sources, MyFixit and X-WLP, at 8 registered discrete settings from exact matching through $\tau = 0.50$. It is not a continuous-interval study and it is not a six-source sweep. Accordingly, this section reports only the quantities present in the two permitted frozen records.

The raw component structure is not invariant. Of the 3 audited quantities—component count, largest-component share, and effective components— 0 remain constant across the registered grid for both sources. Table 6 shows the endpoint values. Component counts and effective-component counts decline, while largest-component share moves at the lower grid points. Calling these raw statistics unchanged would therefore contradict the frozen sweep.

Table 6: Endpoint audit of the registered discrete τ grid. Shares are generated as exact component size divided by frozen source units, not reused as ratio inputs after display formatting. Raw component statistics change between exact matching and $\tau = 0.50$; the final column reports the sum of Holm-surviving flips over all frozen fixture rows and grid points for each source.

Source	n_{comp}		top1 share		N_{eff}		Holm flips
	exact	0.50	exact	0.50	exact	0.50	all rows
MyFixit	85,423	78,563	0.001736	0.007001	13292.7542	6279.5639	0
X-WLP	2,222	2,083	0.005364	0.005875	1448.5611	1204.8758	0

The decision-level negative result is narrower. The frozen winner-identity record contains 32 fixture–source–grid rows. It records 2 raw flips but 0 Holm-surviving flips, and its surviving-flips table is empty. Thus "τ inertness" here means that the registered multiplicity-adjusted flip decision does not change on the registered discrete grid. It does not mean that retained-query counts, rank correlations, winner tie sets, or component statistics are identical. The absence of Holm-surviving flips is a negative result under the registered decision rule; it is not an equivalence test and does not establish that τ has zero effect.

Neither permitted source records a six-source qualifier-set trajectory over this grid. We therefore do not claim that the frozen qualifier set is invariant from exact matching through $\tau = 0.50$. Such a claim would require a corresponding result-blind six-source sweep artifact, which is absent from the frozen record used here.

The edge-family contrast is separately visible in the frozen lattice. Table 7 compares the near-duplicate text layer alone (C2) with its union with source-group edges (C5). For both recorded sources, the configurations have different component counts, largest-component shares, and effective-component counts. This is an observed configuration contrast; it does not identify a general causal mechanism.

Table 7: Frozen near-duplicate edge-family comparison. C2 uses text-near edges alone; C5 uses their union with source-group edges. These are observed configuration contrasts, not a general causal claim.

Source	n_{comp}		top1 share		N_{eff}	
	C2	C5	C2	C5	C2	C5
MyFixit	84,777	5,375	0.001736	0.416427	12870.890	5.760
X-WLP	2,213	180	0.005364	0.104470	1422.480	49.318

On the one outcome reported in both tables, MyFixit's largest-component share, the exact-to-$\tau = 0.50$ endpoint ratio is 4.03, whereas the C2-to-C5 ratio is 239.83; the latter divided by the former is 59.48. This is a descriptive magnitude comparison on a single statistic, not an effect-size test or a significance comparison, and it is not extrapolated beyond the registered grid or this frozen panel. All three ratios are computed from exact integer component sizes rather than the displayed shares. The Gate-C exact endpoint and the lattice C2 configuration happen to share that size. The same largest-component size does not make the two configurations identical.

Within this frozen design, C2 and C5 differ substantially in the audited component summaries, whereas the registered τ sweep yields no Holm-surviving winner flips. These observations concern different outcome levels and are not interpreted as a direct effect-size comparison. The C2/C5 difference is an observed configuration contrast without a general causal interpretation. It does not imply that edge family is universally decisive. It does not imply that a near-duplicate threshold is generally unimportant, that values outside the registered grid are inert, or that the result transfers to other corpora or other near-duplicate methods.

7 A coverage-exposure near-identity

The registered naive-negative exposure is the fraction of all pairs that a naive policy would call negative but that the source policy leaves unresolved. For each corpus, let U be the number of unresolved pairs, R the number of licensed refuted pairs, I the number of incomplete units, and N the total number of units. The two quantities compared here are therefore $U/(R+U)$ and I/N.

For MyFixit, the source-recorded counts give

$$\frac{12,309,728}{14,582,229} = 0.844160, \qquad \frac{198,544}{235,549} = 0.842899,$$

an absolute difference of 0.001261, or 0.126 percentage points. For X-WLP, the same arithmetic gives

$$\frac{27,420}{45,350} = 0.604631, \qquad \frac{2,285}{3,915} = 0.583653,$$

an absolute difference of 0.020978, or 2.098 percentage points.

Table 8: Coverage exposure and the unit-level incompleteness fraction under the one registered exposure definition. Gap is the absolute difference in percentage points; the final columns show the corresponding candidate-pair mass per complete and incomplete unit.

Corpus	Exposure	Incomplete	Gap (pp)	Pairs/complete	Pairs/incomplete
MyFixit	0.844160	0.842899	0.126	61.4	62.0
X-WLP	0.604631	0.583653	2.098	11.0	12.0

The near identity follows from the pair weighting in these two records. The candidate-universe mass is 61.4 licensed refuted pairs per complete MyFixit unit and 62.0 unresolved pairs per incomplete unit. The corresponding X-WLP values are 11.0 and 12.0. Because those complete- and incomplete-unit weights are close within each corpus, weighting units by their candidate-pair mass changes the incompleteness fraction only slightly.

This is an arithmetic near-identity observed on 2 corpora under one exposure definition. It does not establish the same relationship for another exposure definition or another corpus, and it does not establish that coverage-exposure statistics are uninformative beyond these two cases. On these two corpora, the registered statistic carries approximately the same information as the annotated-field completeness rate. This observation is not a critique of any published work.

8 Structured-field unresolved status does not certify lexical silence

E1 assigns `UNRESOLVED` when a structured field lies outside the frozen completeness policy or cannot license negative entailment. That open-world status prevents an unsupported negative label; it is not a certificate that attached natural-language text lacks relevant vocabulary. The distinction is evaluated here only as a deterministic lexical lower-bound diagnostic over the 2 corpora represented in the sole permitted aggregate report.

Table 9: Deterministic lexical lower-bound diagnostic reproduced arithmetically from the hash-verified aggregate report. Positive counts identify frozen-vocabulary cues; they are not semantic labels or requested-value-specific mention rates.

Corpus	Diagnostic	Unresolved	Positive	Rate
MyFixit	Step text: exact normalized frozen-tool cue	198,544	75,247	37.90%
MyFixit	Guide title, subject, or toolbox: exact normalized frozen-tool cue	198,544	195,027	98.23%
MyFixit	Structured tools field intersects frozen vocabulary	198,544	88,757	44.70%
X-WLP	Operation sentence: frozen location-family lexical cue	2,285	1,137	49.76%

The MyFixit step-text predicate is "contains at least one exact normalized cue from the frozen tool vocabulary." Under that predicate, 75,247 of 198,544 unresolved units (37.90%) contain at least one exact normalized cue from the frozen tool vocabulary. The broader guide-title, subject, or toolbox context produces 195,027 (98.23%), while the structured tools field produces 88,757 (44.70%). For X-WLP, 1,137 of 2,285 unresolved operations (49.76%) have an operation sentence that contains a frozen location-family lexical cue. Reporting every frozen diagnostic prevents the broad-context rate from standing alone.

Any positive count is enough to reject the blanket implication that E1 `UNRESOLVED` entails absence of every frozen-vocabulary lexical cue. The converse does not follow. These matches are not semantic labels; a cue need not establish the exact requested relation, and a missing cue does not certify natural-language or semantic silence. Thus structured-field unresolved status does not certify lexical silence, but this audit does not establish semantic sufficiency.

The aggregate report also records requested-value degeneracy. Every MyFixit unresolved replay assertion uses the same universe[0] tool, and every X-WLP unresolved replay assertion uses the same universe[0] location family. The displayed rates therefore cannot be interpreted as requested-value-specific mention rates or as coverage over a representative requested-value distribution.

This section is an aggregate-report arithmetic reproduction from one hash-verified source document, not a raw-corpus independent recount or an end-to-end reproduction. Its scope is limited to these corpora, the frozen E1 construction, and the two recorded relation vocabularies. It does not extend to other corpora, relations, fields, or unresolved definitions, and it is not a critique of published work.

9 Related work and position of this measurement

9.1 Split feasibility and leakage-aware partitioning

Similarity-aware and leakage-reduced partitioning form a mature line of work in biomolecular machine learning. Lo-Hi shows how a giant component in a single molecular-similarity graph can prevent its required split and formulates constrained vertex removal to recover a feasible partition [12]. GraphPart instead uses restricted single-linkage partitioning over a sequence-homology graph, with iterative sequence movement or removal to trade retention against separation [13]. Refnd constructs a thresholded proximity graph, splits at connected components, and optionally subdivides dense components with Leiden before post-filtering [9]. These works give distinct graph-based responses to split feasibility; they should not be read as instances of one identical construction.

Other methods combine several similarity constraints without performing the present layer-versus-union audit. DataSAIL clusters typed entities and optimizes their fold assignments to reduce cross-fold similarity; conflicting assignments in a two-dimensional setting can require interaction removal [8]. LP-PDBBind jointly controls protein-sequence and ligand similarity across train, validation, and test partitions and discards remaining cross-category conflicts before retraining scoring functions [10]. PLINDER constructs metric-specific protein, pocket, protein–ligand-interaction, and ligand graphs, records their connected components and Louvain communities, and permits split configurations that use several graphs and neighbor depths [4]. Together, these studies establish a substantial prior literature on similarity-aware partition design and its retention costs.

9.2 Transitive closure, entity matching, and component formation

The propagation risk of closure also predates the present measurement. In record linkage, Gu et al. describe combining independent matching passes by transitive closure and warn that a false-positive match can propagate across passes [6]. In knowledge-graph entity matching, Baas, Dastani, and Feelders compare closure with weighted cluster editing, which adds and deletes candidate links to enforce transitivity [1]. This literature supplies the relevant closure and repair background, but its evaluated outcome is entity-matching quality rather than component-disjoint benchmark construction across a corpus panel. Classical random-graph theory derives giant-component phase-transition conditions for arbitrary-degree models, including bipartite graphs [11]. We cite this result only as background on giant-component formation; the deterministic typed-layer union audited here is not a test of that random-graph model.

9.3 Near-duplicate and incomplete-judgment evaluation

Information-retrieval evaluation provides a separate adjacent line. Froebe et al. reproduce and extend a study of content-equivalent documents across TREC Terabyte, Web, and Core tracks, showing that duplicate handling can change system evaluation [5]. Buckley and Voorhees introduce bpref to compare retrieval systems when relevance judgments are incomplete [2]. We use these works to delimit evaluation concerns around redundancy and missing judgments; neither is evidence for a typed relation-union split construction.

9.4 Position of the cross-domain measurement

The central direct comparator identified in the bounded search is Guvenilir and Doğan's DTI study [7]. In that setting, the authors construct a heterogeneous network from 3 relation types—compound similarity, protein similarity, and compound–target interactions—compute connected components, and report that a giant component prevents component-disjoint splitting. Their response uses Louvain-guided edge and node deletion. The reported study covers 10 protein-family datasets, 3 split variants per family, and 600 trained models. We therefore attribute the merged-relation, giant-component, split-infeasibility, and deletion/community-repair chain to that work.

That ownership boundary is reinforced by other partial direct precedents. The PPI leakage study groups a graph whose edges encode observed protein interactions or shared sequence clusters before measuring residual interface-structure leakage [3]; PLINDER supports multi-graph split configurations [4]. Among the works reviewed here, we did not identify in the reported designs the conjunction used by this study: one common operational Gate applied symmetrically to individual edge families and their union, individual-layer passes followed by union failure, and a frozen cross-domain procedural-corpus panel. This is a bounded-search positioning statement, not an assertion about unreviewed literature.

The narrower object of the present measurement is therefore the symmetric layer/union audit and its frozen panel outcome. The 2/6 result is a descriptive panel fraction, not a population-frequency estimate. The analysis also reports component-based split-capacity indicators and retains the registered negative result from Section 5: the bridge-specific predictor was not distinguished from the union-density control. This paper contributes neither a splitting algorithm nor an entity-resolution or Louvain/community repair method; it offers no giant-component theorem or causal validation of bridging. We also do not claim priority for observing that merged relations can produce a giant component that obstructs a component-disjoint split.

10 Limitations and scope

10.1 Panel and operational-rule scope

The evaluation panel is small, outcome-enriched, and not drawn at random from a sampling frame for procedural corpora. MyFixit was already known to satisfy the source-level rule, while the remaining 5 sources were subsequently screened candidates. Doc2Dial was the only additional qualifier found in that screening. The resulting 2/6 value is therefore a descriptive fraction of this 6-source panel, not an estimate from independent trials or a prevalence estimate. The panel spans several provenance families and domains, but neither its composition nor its selection process supports a population-level frequency claim.

Eligibility is structural rather than a claim of genre homogeneity. Doc2Dial contains task-oriented government-service documents, whereas other panel members include repair guides, instructional articles, procedure paragraphs, and laboratory protocols. It meets the formal unit, container, and content- field requirements, but that does not make the sources genre-equivalent. Panel heterogeneity therefore limits external interpretation without changing the fixed eligibility decision.

The requirement of at least two qualifying sources is an operational replication rule rather than a statistical replication criterion. Because MyFixit was the known qualifier, its practical application required at least one further qualifier among the five screened candidates. Doc2Dial met that additional-source requirement. This is neither a set of independent repeated trials nor a basis for population-frequency inference.

The source-level decision rule is likewise an operational audit, not a general theorem of split feasibility. A1, A2, and A4 are nested thresholds on largest-component share, whereas A3 uses effective components. Their count therefore weights largest-component concentration repeatedly; the 4 outcomes are not statistically independent. The reserve behind A3 is a prespecified stress margin, not an externally established constant. Moreover, the complete separation produced by A2–A4 in this panel is a post hoc presentation check: it leaves the original A1–A4 rule unchanged and does not show that the same indicators are necessary or sufficient for useful splits elsewhere.

Simple container-size summaries do not reproduce the qualifying set. X-WLP has a larger mean container size than MyFixit (14.03 versus 12.40) but does not qualify. Human Know-How has a larger maximum container size than MyFixit (206 versus 137) and also does not qualify. Doc2Dial's mean and maximum container sizes (64.76 and 494) are comparatively large, so container scale may contribute to its collapse magnitude. The narrower result is only that these simple summaries, by themselves, do not separate both qualifiers from all nonqualifiers; it does not establish that container size is irrelevant or cannot contribute.

10.2 Mechanism and sensitivity limits

The observed association does not identify a bridge-specific mechanism. `bridge_edge_density` and the registered union-density control share union-edge information and are strongly rank-aligned in the 6-source panel. Because the control satisfies the registered separation and outcome-correlation conditions, the mechanism-discrimination requirement is not met. The one-source-deletion separability pattern adds no predictive evidence: strict full-panel separation and retention of both outcome classes make that persistence mechanical. A stronger design would audit predictor geometry before observing outcomes and recruit result-blind sources on which the candidate mechanism and a density control make discordant predictions. The present panel supplies neither a defensible required sample size nor evidence that simple density is itself causal.

The density control is not representation-invariant, although a post hoc spanning-forest sensitivity bound yields the same qualitative negative-control outcome within this panel. The original control is defined on the C3 clique expansion and remains unchanged. The aggregate-compatible bounds in Section 5 preserve qualifier/nonqualifier separation and keep every compatible rank correlation above the registered threshold, but they are panel-specific conservative intervals rather than exact values for a chosen forest. They neither prove representation invariance nor reopen the frozen mechanism decision.

The secondary analyses constrain other interpretations but have narrower domains. Section 6 covers 2 sources and 8 registered discrete settings from exact matching through $\tau = 0.50$; it is neither a continuous-interval analysis nor a six-source qualifier sweep. Raw component summaries change, while 0 winner flips survive the registered Holm decision, so decision-level inertness is not metric invariance or an equivalence test. Section 7 covers 2 corpora under one exposure definition, where pair weighting makes exposure approximately re-express annotated-field incompleteness; it does not extend that arithmetic relation to other definitions or corpora. Section 8 also covers only 2 corpora and reproduces arithmetic from one frozen aggregate report rather than recounting the raw corpora. Its lexical cues are not semantic labels, cue absence does not establish semantic silence, and requested-value degeneracy limits the reported rates. Together these audits bound the findings rather than establishing general threshold, exposure, or vocabulary behavior.

10.3 Evidence and reproducibility boundary

The numerical claims are supported by frozen aggregate artifacts, and separate consistency checks connect those records to the generated tables and manuscript. That closure is not a complete raw-data-to-paper reproduction. The recovered project lacks the complete raw corpora, inherited protocol tree, locked environment, and a one-command path from raw inputs to the reported panel; consequently, the current package supports inspection and aggregate-level recomputation rather than a trusted public runner for the original pipeline.

One unresolved parser risk illustrates the distinction. The recovered Human Know-How loader passes UTF-8 Turtle label bytes through `unicode_escape`, which can introduce mojibake before text normalization and thereby affect content-near-duplicate edges and C2/C5 component structure. No corrected-parser rerun is available, so the direction and magnitude of any change to the frozen Human Know-How metrics cannot be quantified. Current evidence does not require changing the frozen qualifier set, but it also cannot establish that a corrected parser would leave those metrics or that set unchanged. Raw-data reproduction and release of a trusted runner therefore remain separate from the aggregate evidence used here.

10.4 Prior-art and contribution scope

The bounded search is not exhaustive literature proof, implies no criticism of any research community, and cannot establish the absence of an unreviewed comparator. Guvenilir and Doğan previously established the merged-relation, giant-component, split-infeasibility, and deletion/community-repair chain reviewed in Section 9. This paper offers neither a first-observation claim nor a new split algorithm, Louvain/community repair method, giant-component theorem, or causal bridge mechanism. Its retained object is the symmetric comparison of typed layers with their union across a fixed cross-domain procedural-corpus panel, together with negative findings and explicit evidence limits. That measurement remains interpretable within the stated panel and operational definitions but does not establish broader generality.

11 Reproducibility and artifact scope

11.1 Separated verification objects

The public materials separate two verification objects. The companion aggregate artifact contains the frozen records supporting the reported measurements, the scripts used to regenerate the scientific tables and Figure 1, and separate consistency checks for those aggregate-derived outputs.

Existing integrity records apply to those frozen records and generated scientific assets. They do not automatically cover the later Abstract, Introduction, Conclusion, or other editorial prose.

The final manuscript source package contains the final TeX source, bibliography, compilation figures and tables, and the panel-summary and representation-sensitivity CSV files with their deterministic generation script. It is checked through clean extraction, dependency presence, citation resolution, compilation, derived-file byte-for-byte comparison, and PDF font preflight rather than claimed to be covered by the earlier aggregate integrity records. The manuscript-level compile check is distinct from the existing byte-identity records for the frozen aggregate evidence.

When the manuscript source package is paired with the companion aggregate artifact, a clean-directory check rebuilds the aggregate-derived outputs, the editorial panel summary, and the representation sensitivity files. It compares the rebuilt files byte-for-byte with the distributed versions and compiles the manuscript. In operational terms, this checks that the declared aggregate and manuscript dependencies are present, rebuilds the aggregate-derived outputs, and compiles the supplied manuscript source in a clean directory. The figure is a native-LaTeX vector artifact whose plotted inputs are drawn from the frozen lattice and its reconciled table; its source-level outcome is also distinguishable by line weight and marker form rather than color alone.

These checks support aggregate-level inspection and a clean-directory compile of the manuscript package. They do not constitute a raw-data end-to-end reproduction, re-certify extraction of any source corpus, or recover the environment in which the original corpus-wide run was performed. The environment record therefore describes the aggregate check only and does not promise cross-platform byte identity.

11.2 Raw-pipeline and parser boundary

The raw corpora are not part of the public materials. Nor does the recovered project contain the complete inherited protocol tree, the original locked execution environment, a complete inventory of raw inputs, or a one-command path from those inputs to the paper. Consequently, the release does not provide a complete raw-data rerun. The aggregate evidence is preserved as a frozen historical record, while the distinction between that record and a future raw rerun remains explicit.

One recovered Human Know-How loader also has an unresolved decoding risk. It applies `unicode_escape` while parsing labels, which can introduce mojibake before text normalization. Affected normalized text could in turn change content-duplicate edges and the component structure under the content-only and union configurations. No corrected-parser raw rerun is available, so neither the direction nor the magnitude of possible drift in the frozen Human Know-How metrics is known. The current evidence therefore neither establishes that the parser changed the result nor proves that a corrected parser would preserve it. This unresolved path is also why the recovered raw runner is retained only as historical audit code and is not presented as a trusted public execution entry point. Section 10 states the corresponding interpretive limit on the frozen qualifier set.

11.3 Public contents and data access

The companion aggregate artifact is publicly available in the HieraRepair GitHub repository. The public materials consist of two parts. The companion aggregate artifact provides the frozen aggregate evidence, generated scientific tables and Figure 1, generation and validation scripts, and dependency records for aggregate inspection. Its existing integrity records apply to those frozen records and generated scientific assets. The final manuscript source package provides the final TeX source, bibliography, figures and tables, and the two deterministic editorial derivatives with their generator. It is checked by clean extraction, dependency presence, compilation, citation resolution, font preflight, and derived-file byte-for-byte comparison; it is not described as covered by the earlier aggregate integrity records.

Third-party raw corpora are not redistributed. Their acquisition and use remain governed by the original sources and their respective license terms; where the current project record does not verify a license or acquisition path, that status remains unresolved rather than inferred from repository availability. Retrieved literature PDFs and extracted full text used during the bounded prior-art review are likewise excluded. Historical and superseded review materials are not included in the final source archive. Public literature records retain only bibliographic metadata, source identity, locators, and the evidence boundaries needed to audit manuscript citations.

This separation makes the availability statement testable. A reader can inspect the aggregate evidence chain and compile the manuscript from its source package, but must obtain third-party corpora independently for any future raw-level implementation. Neither public component substitutes for a complete raw-input inventory, a trusted raw pipeline, or a corrected-parser rerun; the Human Know-How parser risk therefore remains open.

12 Conclusion

Prior work has already shown that merging relation types can form a giant component, obstruct component-disjoint splitting, and motivate deletion or community-based repair [7]. This study narrows the question to a symmetric measurement across hierarchical procedural corpora: what component structure is observed when a content-near-duplicate layer and a common-container layer are evaluated separately and then combined under transitive closure?

Under the operational source-level rule, 2 of 6 gate-eligible sources qualify. MyFixit and Doc2Dial each retain comparatively dispersed component structure in the individual layers but become highly concentrated under their union. The remaining sources serve as explicit negative cases, including WIQA, for which union formation does not produce the same collapse pattern. The 2/6 result is a descriptive fraction of an outcome-enriched, non-random panel; it is neither a population estimate nor evidence that every relation union behaves similarly.

The mechanism analysis sets a separate boundary. A bridge-specific predictor is associated with the qualifier pattern, but the registered `mean_union_degree` control satisfies the same separation and outcome-association conditions. The panel therefore does not identify a bridge-specific mechanism. The post hoc representation-sensitivity bounds do not change that conclusion: they preserve the qualitative negative-control outcome within this panel while also showing that mean degree itself depends on the clique representation. Likewise, the discrete τ sweep gives a negative result under its multiplicity-adjusted decision rule, not an equivalence result or evidence of a zero threshold effect.

The accompanying diagnostics remain local to their recorded evidence. The coverage-exposure quantity approximately re-expresses annotated-field incompleteness in its two audited corpora because candidate-pair weights are similar across complete and incomplete units. Structured unresolved status does not imply absence of every frozen-vocabulary cue, but lexical matches are not semantic labels. Aggregate records support inspection and recomputation of the reported quantities; the recovered materials do not provide a complete raw-data-to-paper rerun, and the Human Know-How decoding risk remains unresolved.

These boundaries define the paper as a measurement and audit rather than a new splitting method or a general causal account. Future studies can improve mechanism discrimination by assessing predictor geometry before outcome testing and recruiting result-blind sources on which candidate and control predictors disagree. Within this panel, layer-wise acceptability was not sufficient evidence of union-level capacity; the union therefore merits audit as an object in its own right.

AI assistance disclosure

OpenAI ChatGPT/Codex and Anthropic Claude were used to assist with code and evidence auditing, literature organization, and manuscript drafting and editing. The authors reviewed the resulting analyses, citations, and text and take full responsibility for the final content.

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